

Arbaminch University
Institute Of Technology
Hydraulic And Water Resources Engineering Department

Course Name: Engineering Hydrology

Course Code: HENG-3143

Examiner: Assefa.G & Wondafrash A.

Section: G3HWRE-SEC A, B, C &D

Date: Feb 2/2016

Time Allowed: 3:00hrs

Final Exam

Max. Mark 50%

General Instructions:

- Attempt All The Questions Carefully.
- Neatness Deserves Weightage.
- The Exam Is Closed Book.
- Assume Any Missing Data.

Name _____

ID.NO _____

6571

To be returned along with answer sheet

1. Discuss the following briefly:

- a. The basic objective of flood frequency analysis and its practical application. (3pts)
- b. The importance of flood routing and reservoir capacity determination. (3pts)

2. Explain the following terms

- a. Annual series and partial duration series (2pts)
- b. Recurrence interval (2pts)

3. Discuss the impact of sediment on the useful life of different hydraulic structures and its various methods which are adopted to control the deposition of sediment in reservoir. (3pts)

4. The record of annual peak discharge at a stream gauging station is as follows

year	1980	1981	1982	1983	1984	1985	1986	1987	1988
Q(m ³ /s)	311	720	430	330	248	125	405	400	233
year	1989	1990	1991	1992					
Q(m ³ /s)	178	248	321	186					

Determine

- a. The magnitude of a 10 year flood by using Log Pearson type III distribution methods. (5pts)
- b. The confidence limit of 95% by using Gumbel estimate for flood discharge of 10 years. (8pts)
- c. Reliability of the project by assuming its expected life is 25 years and designed for a flood magnitude of return period of 50 years. (2pts)

5. For a river reach $k=8h$ and $x=0.25$, route the following inflow hydrograph. (5pts)

Time(h)	0	4	8	12	16	20	24	28
Inflow(m ³ /s)	8	16	30	30	25	20	15	10

The initial outflow discharge from the reach is 8m³/s.

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6. In installation of storm water drainage system for a catchment urban area, there are three pipes AB, BC, CD with the outfall at D. The details of the pipes and the areas they serve are as follow.

Pipe	Diameter(mm)	Length(m)	Gradient (1 in...)	Area(ha)
AB	533	450	225	2.9
BC	610	390	238	2.9
CD	686	225	250	2.4

- Assume a time of entry of 2min and roughness of the pipes (n) is 0.013. determine which pipe length has insufficient capacity to carry a way the 2 year storm for which the intensity $i(\text{mm/h})$ related to duration $t(\text{min})$ by expression $i=825/(t+8)$. (use manning equation to determine the permissible velocity and discharge of each pipes). (7pts)
7. The following inflow and demand volumes (Mm^3) information's are available at a reservoir site located downstream of another reservoir. (10pts)

- a) Calculate the storage needed to meet the demand by sequent algorism method

Month	Inflow Mm^3	Flow from Intermediate catchment Mm^3	Irrigation Demand	Other Demands	Evaporation losses	Suspended sediment in M m^3
Jan	3.2	0.6				
Feb	2.4	0.4	1.3	0.1	0.3	1.2
Mar	2	0.4	1.4	0.1	0.4	0.5
Apr	1.4	0.2	1.5	0.2	0.5	0.2
May	0.5	0.1	1.8	0.2	0.6	0.1
Jun	0.9	0.1	1.3	0.2	0.8	0.12
Jul	15.8	0.9	1.6	0.2	0.8	0.22
Aug	18	1.1	2.5	0.1	0.8	2.0
Sept	13.1	0.8	2.1	0.1	0.7	3.0
Oct	5.2	0.8	1.7	0.1	0.6	1.5
Nov	4.1	0.7	1.0	0.1	0.5	1.2
Dec	3.3	0.5	0.6	0.1	0.4	1.0
			0.8	0.1	0.3	0.8

- b) Depending on the above data develop a regression relation between suspended sediment load and discharge at the site.
- c) Predict the suspended sediment for a discharge of 50Mm^3

0.2923 m^3/s
0.4182 m^3/s

2.

Design a tube well having 100m rad with a velocity of 2.3m/s and drawd below the ground surface in wet seas

Depth, m	0-15	15-25
formation	Surface Clay	Coarse Sand

- ✓ Hydraulic conductivity = 0.03
- ✓ Pipe friction factor = 0.024
- ✓ Length of delivery pipe = 10m
- ✓ Pump efficiency = 65%
- ✓ Affordable pipe dimensions = 5
- ✓ Opening area percentage = 15%

13.

Rainfall at the rate of 6mm/hr falls on two lakes having water depths located respectively. The land is underlain by Take a hydraulic conductivity of 20m rainfall infiltrates into the soil. Compute

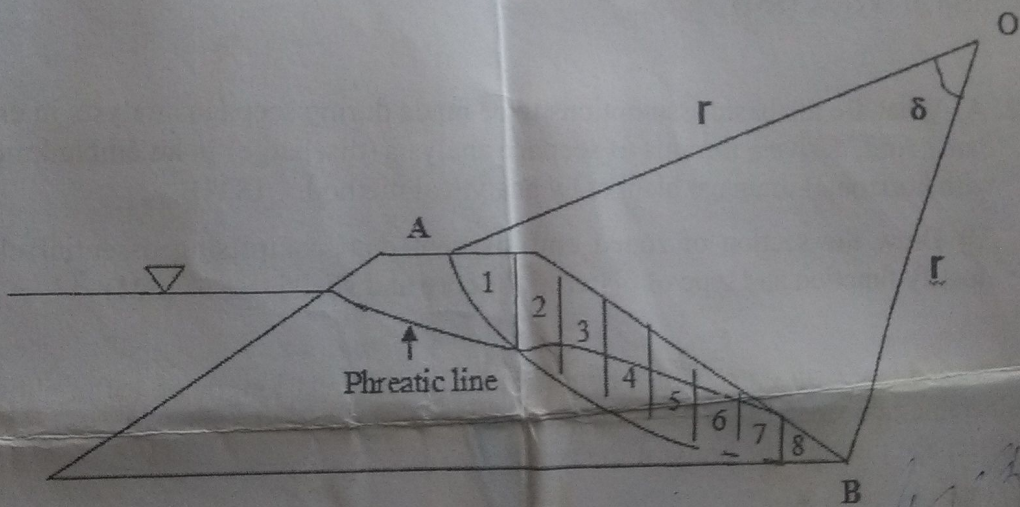
- a. Equation of pheratic surface
- b. Location of divide point
- c. Flow into both lakes
- d. If there is no rainfall what would be the

14.

A 20cm well fully penetrates a 20m de distance of 15m and r_2 m from a pump 0.70m respectively. Calculate the pum distance for both confined and unconfir the pumped well is 3.50m and having a

74

Slice No	Angle θ	Area of slice (m^2)	
		Above Phreatic surface	Below phreatic surface
1	54.5	12.2	0
2	41	15	4.3
3	31	14.3	7
4	22	11	10
5	13	7.5	13
6	5	5	12
7	-3.5	3	9
8	-13	0	7



Have a Nice work!

$$b = 0.3 \times 2.5 \times 8.6 + 1.5 \times 2.4$$

$$(b, w) = (8.9, 2.6)$$

Code: HENG-3144

Time allowed: 3:00hrs.

Semester: I (2015/16 A.C.)

Examiner; Wegayehu A. and Fikirte S.

Instruction:

- ➡ Assume any reasonable data
- ➡ Put the values in a standard
- ➡ Clear steps and neat sketches
- ➡ The exam is closed book.

PART I

1. Define and briefly enumerate the following pair of terms.....

- a. Observation Well and Tube Well
- b. Actual Drawdown and Residual Drawdown
- c. Radius of the Well and Radius of Influence
- d. Electrical Resistivity and Seismicity
- e. Well Loss and Specific Capacity

2. Elaborate the development assumption in the well hydraulics.....

3. ABC international investment group has convinced government officials to prospect for a bottling factory. If the company has a geology team, what a series of engineering potential groundwater site for the

PART II

1. For well fully penetrating a confined aquifer

**ARBAMINCH UNIVERSITY
INSTITUTE OF TECHNOLOGY**

HYDRAULIC & WATER RESOURCES ENGINEERING DEPARTMENT

Course Name: Hydraulic structure-I
Course code: HE-3151
Examiner: Girma G.
General Instructions:

Final Exam
MAX.MARK 50%

Section: G₂HWRE-A&B
Date: 28/01/2017
Time allowed: 3:00hrs

- > Attempt all the questions carefully.
- > Neatness deserves weightage.
- > The exam is closed Book.
- > Assume any missing data.

Name _____
ID.No _____
To be returned _____

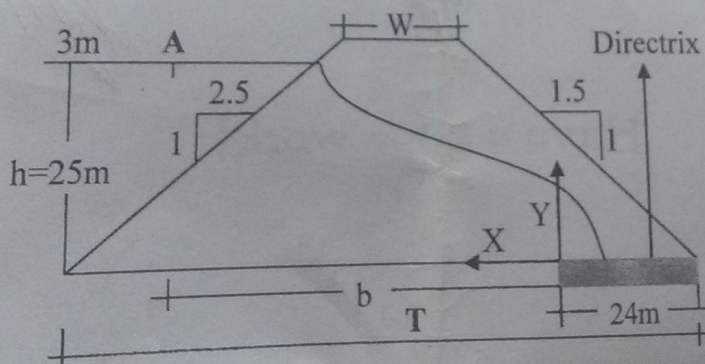
Q#1.A). What are the causes of failures in earth fill dam? And, how do we avoid it? (5M)
B) State and justify in detail the two purposes of river diversion takes places with the help of sketch. (5M)

Q#2. A) What are the basic assumptions to be made during seepage analysis in embankment dam? And, derive a formula of seepage analysis (discharge) in an embankment dam with horizontal drainage blanket by analytical method. (8M)
B) Draw the section of zoned embankment dam comprising essential elements and justify function and type of material for core and shell? (5M)

Q#3. A medium earth dam with a height of 28m and crest length of 150m shown below is vulnerable to seepage failure. Find the crest width, locate the phreatic line and determine the seepage through the body of the dam?

Take $k=3 \times 10^{-6} \text{ m/s}$

Choose: (12M)



$$W = \frac{H_d}{5} + 3$$

$$W = 1.65(H_d + 1.5)^{1/3}$$

$$W = 0.55\sqrt{H_d} + 0.20H_d$$

Q#4 Determine the factor of safety for the slip surface on the downstream portion of the dam under steady seepage condition as shown below for the following data.

Saturated unit weight of soil = 22 KN/m^3 , radius of entire slip surface, $r=153\text{m}$
Moist unit weight of soil = 18 KN/m^3 , central angle of slip surface, $\delta=30^\circ$
Unit weight of water = 10 KN/m^3
Unit cohesion = 24 KN/m^2
Angle of internal friction, $\Phi = 30^\circ$

(15M)

Total pressure (s) = 202

Hydraulic and Water Resources Engineering Dep't
Water treatment. Final exam for 3rd year HWRE students.
Time allowed: 3hrs

C580

Part-I discuss briefly the following questions

1. What is the conventional method to treat water? (5pts)
2. What is coagulation, what is removed during coagulation? What is added? (3pts)
3. What happens to water during filtration, what is removed during filtration? (4pts)
4. Write the steps to treat surface water in water treatment processes. (3pts)
5. Discuss the theory of filtration, types of filtration and also the difference, advantage and disadvantage of slow and rapid sand filter. (3pts)
6. State the factors that affect sedimentation. (2pts)
7. What are the points to be considered for locating a treatment plant? (3pts)

Part-II solve the following questions

1. 2 million liters of water per day passing through sedimentation tank which is 6m wide, 15m long and having a water depth of 3m. Find (8pts)
A) Detention period
B) Average flow velocity
C) Amount per day of suspended solid if the contamination is 60ppm and sp. gravity is 2. Assume 70% removal in the tank
D) Overflow rate
2. Design a coagulation sedimentation tank with continuous flow for populations of 80,000 people with daily per capita water demand of 120 liters. Assuming a detention period of 6hr, a width of 12m, overflow rate of 1500lit/m²/hr, and also make suitable assumption if needed. (3pts)
3. Design a slow sand filter bed for the following data: Population → 50,000, Per capita demand → 150lit/person/day, Rate of filtration → 180lit/hr/m², Length of bed twice its width, Assume 1 unit out of 6 as standby. (3pts)
4. Design a sedimentation tank for a purification plant of max. Daily demand of 12 Million litres. Assume 6 hr. detention period and 20 cm/min. flow velocity. Assume a depth of 4m, Assume 0.50m freeboard. (5pts)
5. determine the terminal settling velocity for a sand particle with an average diameter of 0.5mm and a density of 2600kg/m³ density and viscosity of water at 20degree centigrade is 998.2 and 10^{-3} N.s/m². (3pts)
6. Find out the quantity of alum required to treat 18million litres of water per day. The dosage of alum is 14mg/litre. Also work out the amount of CO₂ released per litre of treated water. (3pts)

Good Luck

Prepared By: Water supply & urban drainage Engineering Stream.

$$2 \times 10^6 \times \frac{60}{1000} = 120$$

Part I. Read the following statements and write T if the statement is CORRECT, F if it is INCORRECT on the space provided. (10 pts)

1. We can understand engineering as science better if we look at the relationship between science and technology in general. T
2. According to classical view of the analysis of engineering and science, the relationship between science and technology is linear, first science then technology. T
3. According to classical view of the analysis of the relationship between science and technology, science, technology and society play role. F
4. What differentiates a report from other types of writing is it is a form of professional documentation designed primarily to be selectively read section by section. T
5. In technical writing it is possible to place large and complex illustrations in the Appendix Section. T
6. List of Illustrations section is titled as List of Tables if the report contains tables only. T
7. According to the convention of numbering pages, all preliminary pages are numbered in lower case Roman numbers. F
8. The purpose of environmental reports in general is to assess the conditions, uses and significances of a site or ecosystem. T
9. Design reports are different from other types of writing for they are open-ended, require creativity and have no any set of rules. F
10. Technical writing is one of the major steps when conducting research (planning, gathering, analyzing and interpreting data). T

Part II. Match each section of a report in Column "A" with their explanations with Column "B". (10 pts)

Column "A"

1. Title & Cover Page
2. Summary
3. Acknowledgments
4. List of Symbols
5. Introduction
6. Theory
7. Middle Sections
8. Conclusions
9. References
10. Appendices

Column "B"

- A. as important as summary to understand the work drawn directly and logically from the findings.
- B. conventional way of listing sources of information used for the study.
- C. detailed material that would otherwise clog up the text in the body of the report.
- D. gist of the whole report including the most important parts of the report.
- E. explains works and literatures behind the report, not including too much.
- F. undecorated, normal and formal, informative not too long or too short.
- G. gives credits to the supervisor, members of staff, people in the industry or other organizations
- H. defines key terms used in the report including technical terms, symbols, abbreviations, etc.
- I. describes background of the work, starting with the broad context and leading up to the hypotheses.
- J. most important section which varies from field to field, but logically sequenced headings

I. Read the following statements and write TRUE if the statement is CORRECT, FALSE if it is INCORRECT on the space provided. (11 pts)

- True 1. When summary is written it is not important to mention specific figures or to cite references.
- False 2. According to the use of illustrations, linear line graphs are more appropriate to represent separate parts of a whole than pie charts.
- T 3. We need tables when we want to show classifications.
- T 4. According to the convention of numbering pages, all the preliminary pages are numbered in lower case Roman numbers.
- F 5. Field trip reports are simply regurgitations of general material by the observer *our observations*
- T 6. Every figure and table must be referred to somewhere in the text at a relevant point in the text either at the beginning or at the end of a sentence.
- OTV 7. It is possible to leave illustrations unnumbered and unlabelled sometimes referring to in the text as '... as shown in the illustration below'.
- F 8. Writing in the report something like, 'The data obtained are given in Appendix 3' and leading the reader to the appendix section is the conventional ways of technical writing.
- True 9. The notion of citation and referencing is that every time we use other's ideas we have to acknowledge them conventionally and scientifically *others*
- T 10. According to the convention of citation and referencing, we use references when we cite factual materials from other sources.
- F 11. Acknowledging diagrams is not mandatory even if they are taken from another source.

II. Chose the correct answer and circle the letter of your choice. (7 pts)

1. One of the following is NOT one of the major uses of illustrations.
- A. helping the reader understand the report clearly
 - B. making the report look professional
 - C. enabling the writer create visual images on the reader
 - D enabling the writer select concepts from sources
2. What are the points we should consider to decide which illustration to use to which report.
- A. How are the most important points in the data best classified?
 - B. How much detail and accuracy are required?
 - C. Which source material is most appropriate?
 - D A & B are correct
 - E. B & C are correct

Paraphratic

3. Which one of the following is FALSE about the guidelines for numbering illustrations and referring to them in the text.

A. Illustrations are numbered in the way that you deal with them in the text.

B. There should be two numbering series- one for figures one for tables.

☒ C. There should be one numbering series for both figures and tables.

D. Even if your illustration are maps, graphs and/or line drawings, call them Figure.

4 The two major aspects we should consider when we document sources are ✓

A. Citation and Harvard System

B. Harvard System and Numbering System

☒ C. In-text Citation and List of References

D. In-text Citation and Use of Illustrations

☒ 5 The disadvantages of the sequential numbering system is

A. It is difficult to add another citation and renumber all successive ones without getting a muddle.

B. The numbers give no information about the work, and it is easy to forget to use the earlier number to refer to it again. (easy)

C. A reader familiar with the literature cannot immediately recognize the piece of work you are citing. (done)

D/ All are answers E. No correct answer is given

5. Environmental reports: The purpose of this report is to report on the condition, use and significance of a site or ecosystem. There are two different forms of an environmental report. 1) an environmental impact report and 2) an environmental assessment report.

The purpose of an environmental impact report is to assess the impact of a proposed development or change in use. The purpose of an environmental assessment report is to provide basic information on a natural environment for the planning department of the local authority. It also provides an objective and subjective assessment of the value of the site or ecosystem to encourage protection of the area.

Suggested structure for these reports would be:

Summary
Introduction
Description of the area - its locality, geography, and present condition
Assessment of the value - as an amenity, natural system, aesthetic feature etc requiring comparison with others nearby.
Analysis of its significance, urgency of action, irreversibility of action and extent of loss to society or community if not protected.

For an impact report: An account of the impact of the development, and the environmental safeguards to be adopted.

Recommendations, Conclusions, References, Appendices

6. A field trip report: the purpose of this report is to report on a field investigation. In the biological sciences, they generally describe and in the earth sciences with aspects of geological formations and fossil life. It is essential that these reports should not be a duplication of general material, but should present own observations.

In Biological sciences, reports are usually required in the standard format as follows.
Summary, Introduction, Theory (if needed), Experimental procedure, results, discussion, conclusions, references, appendices. - *investigation first hand information*

In Earth sciences, the suggested headings would be: summary, introduction, land forms, Lithology and Faunas, structure, discussion, references and appendices.

7. Design Reports: Engineering / software development: The purpose of the report is to give the specifications for an engineering design, or for software development. To design is to create the description of something that does not yet exist. Therefore, these reports are quite different from other types. These reports are: open ended, require creativity and individual judgement, and must satisfy a set of rules.

It is essential to deliver the ideas clearly. There is no general format for engineering design reports, since it will depend on the individual departments and lectures. Some departments regard the workbook as the design report; others require an assembling of the material into something that closely resembles other type of report. It is essential to conform to the requirements of the department. The following general points are to be noted for clear presentation of the report.

1. The calculations or coding must be presented in such a way that someone can follow them. Anticipate the way in which marker is most clearly going to understand the reasoning, and then present the sequence of design.
2. Drawings must be self explanatory.
3. There is a need to quote and fully reference formulae, standards or codes, and any assumptions made.

Class Activity 1

Structuring a technical report

Name:

ID No

Section

11

You have already got some knowledge of the basic structure/ basic skeleton of a technical report and also got some idea of what each section is about.

Which of the following structures do you think is the best?

Title Page
Contents Page
Theory (if needed)
Introduction/background
Middle section
Conclusions
Recommendations
Glossary of terms
Appendices
References (Bibliography)

Abstract (Summary)
Contents Page
Glossary of Terms
Introduction/ background
Theory (if needed)
Middle sections
conclusions
recommendations
References (Bibliography)
Appendices

Title Page
Contents Page
Theory (if needed)
Introduction/background
Middle Section
Conclusions
Recommendations
Glossary
Appendices
References (Bibliography)

b) Choose a type of technical report (out of 6 given) and write the basic structure

Choose the best Title

- 1) Analysis and Comparison of the Transportation Systems of Several Major Cities
- 2) Analysis of Urban Transportation Systems
- 3) Analysis of Transportation Systems

*Ten Laws of RW/TRW/TC

1. The Reader is the most important person
2. Keep the report as simple as possible
3. organize for the convenience of the report user.
4. All reference should be correct in detail
5. The writing should be accurate, concise and unobtrusive.
6. The right diagram with the right labels should be in the right place for the reader.
7. summaries give the whole picture in miniature.
8. Reports should be checked for technical errors
9. The report should look as good as it is

(4) Writing Research Reports

Producing an outline of a research report

You are now in a position to think about the overall plan of your report. This is known as the skeletal framework. It is like drawing up the plans for a new house. Not only will it show its overall structure, it will also remind you of the materials (information) you will need to gather before the process of construction can begin.

A number of significant benefits will accrue in constructing a skeletal framework. In particular, it will enable the writer:

- to be sure there is no misunderstanding over the Terms of Reference
- to have an overview of the entire report to be reminded of what information must be collected, what is already available and what is not needed to order his or her thoughts before considering how they should be expressed
- to appreciate the significance of, and the relationship between, the materials that will be gathered

Report writing

APPA - Research

1. Read the following questions and give short answers to each in the space provided. Limit your answers to only the space given.

2. Write the most important steps of scientific research. (2.5pts)

1. Define the problem
2. Formulate a hypothesis
3. Design an experiment
4. Conduct the experiment
5. Analyze the results
6. Draw conclusions

3. Write the four goals and characteristics of scientific research. (2pts)

1. Goal: Understanding the world
2. Goal: Prediction of events
3. Goal: Description of events
4. Goal: Control of events

4. What are the important attributes embedded in the definition of engineering as science? (2 pts)

1. Engineering is a science
2. Engineering is a technology
3. Engineering is a design
4. Engineering is a process

5. Research in engineering can be performed rationally by structuring it according to the engineering cycle. Write the steps of the engineering cycle. (2.5pts)

1. Problem Identification
2. Problem Definition
3. Problem Formulation
4. Problem Solution
5. Problem Evaluation

6. Although research methods for other sciences and for engineering sciences and similar, the following are the most commonly used methods in engineering. (2.5pts)

1. Experimental method
2. Theoretical method
3. Survey method
4. Case study method

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Part II. Match each section of a report in Column "A" with their explanation with Column "B". (10 pts)

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2. Summary	B. conventional way of listing sources of information used for the study.
3. Acknowledgments	C. detailed material that would otherwise clog up the text in the body of the report.
4. List of Symbols	D. gist of the whole report including the most important parts of the report.
5. Introduction	E. explains works and literatures behind the report, not including too much.
6. Theory	F. understated, normal and formal, informative not too long or too short.
7. Middle Sections	G. gives credits to the supervisor, members of staff, people in the industry or other organizations.
8. Conclusions	H. defines key terms used in the report including technical terms, symbols, abbreviations, etc.
9. References	I. describes background of the work, starting with the broad context and leading up to the hypothesis.
10. Appendices	J. most important section which varies from field to field, but logically sequential headings.

III. Match types of report in Column "A" with their main characteristics from Column "B". (7 pts)

Column "A"

Column "B"

- F 1. General Report
E 2. Laboratory Report
E 3. Research Project
A 4. Practical Work Report
G 5. Environmental Report
D 6. Field Trip Report
B 7. Engineering Design Report

- A. This is a report on work for an outside organization with the aim of describing the activities of the company and work within it.
 B. This is an open-ended report requiring creativity and individual judgment with a set of rules and with the purpose of describing new innovations.
 C. This report is an original work carried out in a group or as exercise with the aim of documenting and sharing your findings with readers.
 D. This report is on field investigations and observations with the purpose of presenting what is observed.
 E. The report should be within the frame work of the department describing the procedures and presenting the results including a results and discussion.
 F. This report is on a specific area with the purpose of giving a balanced and illustrated account of a particular area of knowledge.
 G. The purpose of this report is to report on the conditions, uses and significances of a site or ecosystem with its two different forms.

IV. Fill in the blank spaces with appropriate answer. (5pts)

1. Reference means acknowledging the sources of information you have used. Bibliography
2. When we use other's ideas we paraphrase (expressing other's idea in our words), summarization (expressing concisely in our words), quote copy (expressing in their exact words), and illustration (reproduce a diagram, table or graph).
3. The term illustrations includes both tables and figures when the report has many graphs, tables, line drawings, etc.
4. Failure to acknowledge the sources is called plagiarism.

Plagiarism

Plagiarism

"Writing Report"

Part-I (2 points for each)

1. Clearly mention what does mean technical report writing
2. Write a short note on the goal of technical report writing and research methods
3. Mention the different between oral and poster presentation
4. Mention clearly with example about deductive and inductive logic
5. List elements of technical report writing and elaborate each element with short example
6. Mention types of sampling design

Part-II

1.0. Make available your research title and based on your research title clearly state your keywords and create your concept map. (6 points)

2.0. From the given reference list place the document citation by APA methods. (6 points)

A. Awulachew, S.D and Mone, J.D (2007). Assessment of Small Scale Irrigation and water harvesting in Ethiopian Agricultural Development, International Water Management Institute (IWMI), Addis Ababa, Ethiopia.

B. Biswas, Asit K (1984). Monitoring and evaluation of an irrigation system, International Journal of Water Resources Development, Vol 2: No1, pp 3-25.

3.0. Using the information provided below write the reference for the document by APA methods. (6 points)

A)

Author name - Tamru Tesseme

Year - 2012

Article Title - Improvement of AMU ground water quality by natural Zeolite for drinking application

Journal Title - 12th Symposium on sustainable water resources development

Publication details and Page - Arba Minch University, VOL-1 page 12-20

Tamru Tesseme (2012). Improvement of AMU ground water quality by natural Zeolite for drinking applo. Journal Title 12th S. , Arba Minch University

Arbaminch University
Institute Of Technology
Hydraulic And Water Resources Engineering Department

6. In installation of storm water drainage system for a catchment urban area, there are three pipes AB, BC, CD with the outfall at D. The details of the pipes and the areas they serve are as follow.

Pipe	Diameter(mm)	Length(m)	Gradient (1 in...)	Area(ha)
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- Assume a time of entry of 2min and roughness of the pipes (n) is 0.013. determine which pipe length has insufficient capacity to carry a way the 2 year storm for which the intensity $i(\text{mm/h})$ related to duration $t(\text{min})$ by expression $i=825/(t+8)$. (use manning equation to determine the permissible velocity and discharge of each pipes). (7pts)
7. The following inflow and demand volumes (Mm³) information's are available at a reservoir site located downstream of another reservoir. (10pts)

- a) Calculate the storage needed to meet the demand by sequent algorism method

Month	Inflow Mm ³	Flow from Intermediate catchment Mm ³	Irrigation Demand	Other Demands	Evaporation losses	Suspended sediment in M m ³
Jan	3.2	0.6				
Feb	2.4	0.4	1.3	0.1	0.3	1.2
Mar	2	0.4	1.4	0.1	0.4	0.5
Apr	1.4	0.2	1.5	0.2	0.5	0.2
May	0.5	0.1	1.8	0.2	0.6	0.1
Jun	0.9	0.1	1.3	0.2	0.8	0.12
Jul	15.8	0.9	1.6	0.2	0.8	0.22
Aug	18	1.1	2.5	0.1	0.8	2.0
Sept	13.1	0.8	2.1	0.1	0.7	3.0
Oct	5.2	0.8	1.7	0.1	0.6	1.5
Nov	4.1	0.7	1.0	0.1	0.5	1.2
Dec	3.3	0.5	0.6	0.1	0.4	1.0
			0.8	0.1	0.3	0.8

- b) Depending on the above data develop a regression relation between suspended sediment load and discharge at the site.
- c) Predict the suspended sediment for a discharge of 50Mm³

0.2923 m³/s
0.4182 m³/s

**ARBA MINCH UNIVERSITY
INSTITUTE OF TECHNOLOGY
HYDRAULIC AND WATER RESOURCES ENGINEERING DEPARTMENT
Final Examination**

Course Name: Hydraulic Structures - I
Course Code: HE- 2152
Examiner: Hydraulic str. team
General instruction:

Section: G2HWRE:-A-H
Date: June/2012
Time allowed: 3hrs
Max. Mark: 50%

- ☞ Attempt all the questions carefully
- ☞ Neatness deserves weightage
- ☞ The exam is closed book
- ☞ Assume any missing data reasonably

3. A) The volume of concrete in a concrete arch element of unit length is given

$$V = \frac{2\gamma Z R_u^2 \theta}{\sigma_{all}}$$

Where: γ : unit weight of water
 Z : depth under the water
 R_u : upstream radius of curvature
 θ : half of arch angle

Show that the minimum volume of concrete is achieved when the value of $2\theta = 133^\circ 34'$ (5points)

B) Design a constant radius arch dam for a valley a 100m, at the base and 150m at a height of 100m use allowable stress of concrete $\sigma_{all} = 4 \text{ MN/m}^2$. Draw the profile of the dam at 10m intervals. (10points)

A) As an engineer explain:- The Various causes and ways of failure of earth dam. (5points)

B) Surface shown for steady seepage with the following data and table below. (10pts)

Average unit weight under steady seepage = 20 KN/m^3

Cohesion, $C = 24 \text{ KN/m}^2$

Angel of internal friction, $\phi = 25^\circ$,

Number of Slices = 9 Determine the factor of safety for the slip.

Slice no.	Degree	Arc length	Area of Slice (m^2)	Pore water pressure
1	55	8.6	45	0
2	50	8.7	49	150
3	45	5.4	51	250
4	41	6.4	80	300
5	32	6.3	87	290
6	23	5.9	54	210
7	15	6.2	52	95
8	12	6.0	48	0
9	0	7.4	44	0

GROUND WATER ENGINEERING FINAL EXAM

Course Title: Ground Water Engineering

Target Group: G3 HWRE

Course Code: HENG 3144

Maximum Mark: 50%

Academic Year 2017

Time Allowed: 3 Hours

Final Examination (3rd Year, 1st Semester)

Examiners: Yosef M. & Aklilu A

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tube
des

Please give your answers as brief as possible.

Part One (Short and Precise Answer)

1. Write short notes on the following (not more than 3 lines for each) (5 points)
 - a. Well screen
 - b. Gravel packing
 - c. Drilling fluid
 - d. Well grouting
 - e. Well development
2. List down ground water resources management issues & explain 5 of them briefly. (4 points)
3. What is meant by artificial recharge of ground water, list down techniques of artificial recharge & explain them briefly. (4 points)
4. Suppose Gamo-Gofa zone water and mineral office needs to conduct pumping test in Beto Woreda for irrigational purpose from ground water wells, while locating test wells to carry out pumping tests, there are considerations that should be taken into account. List down these considerations. And also explain briefly types of pumping test that can be done to determine hydraulic parameters of the well. (5 points)
5. It is required to identify a potential aquifer for public water supply purpose. So explain those stages you need to pass through in your study to locate the perfect site for production well as a hydraulic engineer and an expert in the field. (5 points)
6. Explain the principles behind both electrical resistivity and seismic refraction method in applying them for ground water exploration purpose. (4 points)

used to construct wells by alternately lifting & dropping a set of drilling tools. Surplus
loose material & drill cuttings are raised in drilling water. By the action of the
Direct circulation drilling method this method uses rotary bit to cut the
rock & circulating drilling fluid to flush rock cuttings to the surface.
Circulating drilling fluid to flush rock cuttings to the surface.
usually heavy mud which is able to support the walls

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Hydraulic And Water Resources Engineering Department

6. In installation of storm water drainage system for a catchment urban area, there are three pipes AB, BC, CD with the outfall at D. The details of the pipes and the areas they serve are as follow.

Pipe	Diameter(mm)	Length(m)	Gradient (1 in...)	Area(ha)
AB	533	450	225	2.9
BC	610	390	238	2.9
CD	686	225	250	2.4

- Assume a time of entry of 2min and roughness of the pipes (n) is 0.013. determine which pipe length has insufficient capacity to carry a way the 2 year storm for which the intensity $i(\text{mm/h})$ related to duration $t(\text{min})$ by expression $i=825/(t+8)$. (use manning equation to determine the permissible velocity and discharge of each pipes). (7pts)
7. The following inflow and demand volumes (Mm³) information's are available at a reservoir site located downstream of another reservoir. (10pts)

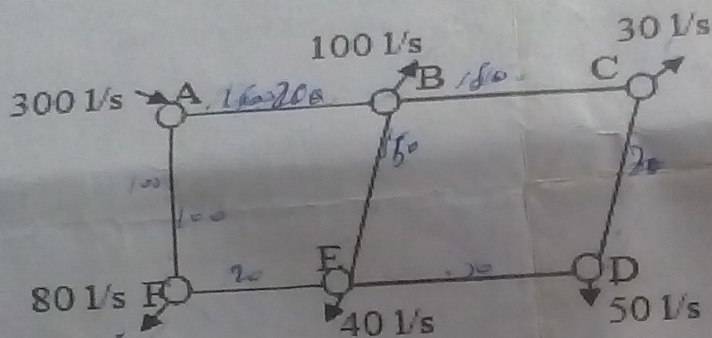
- a) Calculate the storage needed to meet the demand by sequent algorism method

Month	Inflow Mm ³	Flow from Intermediate catchment Mm ³	Irrigation Demand	Other Demands	Evaporation losses	Suspended sediment in M m ³
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- b) Depending on the above data develop a regression relation between suspended sediment load and discharge at the site.
- c) Predict the suspended sediment for a discharge of 50Mm³

0.2923 m³/s
0.4182 m³/s

Final exam



13. A square rapid mixing basin, with a depth of a water equal to five fourth of the width, to be designed for a flow of $1000 \text{ m}^3/\text{d}$. The velocity gradient is to be $600/\text{s}$, the detention time is 0.2 minutes operating temperature is 10°C , and the turbine shaft speed is 100 rpm. Determine:

- The basin dimension (3.5 marks)
- The power required (2.5 marks)

Take: $\mu = 0.0013 \text{ N}\cdot\text{s}/\text{m}^2$ at 10°C

14. It is required to install rapid sand filters downstream of the sedimentation tank. The design loading rate is selected to be $180 \text{ m}^3/(\text{m}^2/\text{d})$. the maximum surface per filter is limited to 40 m^2 . Design the number & size of the filters and calculate the normal filtration rate. (4 marks)

1) 4702 m²

**ARBA MINCH UNIVERSITY
INSTITUTE OF TECHNOLOGY**

**HYDRAULIC AND WATER RESOURCES ENGINEERING DEPARTMENT
Final Examination**

Course Name: Hydraulic Structures – I

Section: G2HWRE:-A-H

Course Code: HE- 2152

Date: June/2012

Examiner: Hydraulic str. team

Time allowed: 3hrs

General instruction:

Max. Mark: 50%

- ☞ Attempt all the questions carefully
- ☞ Neatness deserves weightage
- ☞ The exam is closed book
- ☞ Assume any missing data reasonably

C) A medium earth dam with a height of 28m and crest length of 150m shown below is vulnerable to seepage failure. Find the crest width, locate the phreatic line and determine the total seepage through the body of the dam? Take, $K = 3 \times 10^{-6} \text{ m/s}$,
(Only G2HWRE-A, G&H)

Hint: use analytic method (10pts)

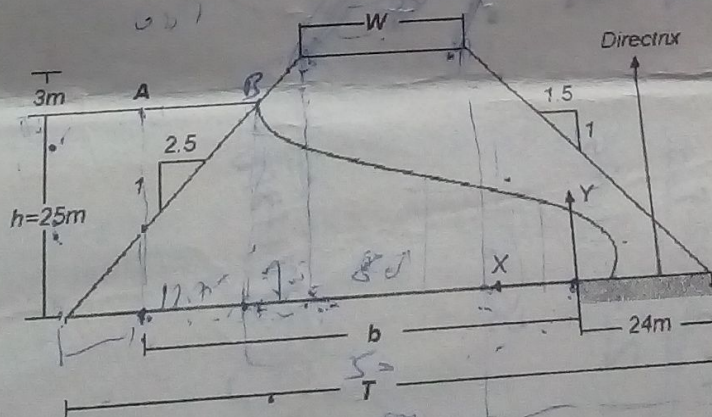
Choose: $W = \frac{H_d}{5} + 3$

$W = 1.65(H_d + 1.5)^{1/3} > 3m$

$W = 0.55\sqrt{H_d} + 0.20H_d < 3m$

$W = 0.55\sqrt{28} + 0.20 \times 28$

$W = 8.11$



4. Explain the meaning, purpose and ways of river diversion with net sketch. (5pts)

Arbaminch University
Institute Of Technology
Hydraulic And Water Resources Engineering Department

Course Name: Engineering Hydrology

Course Code: HENG-3143

Examiner: Assefa.G & Wondafrash A.

Section: G3HWRE-SEC A, B, C &D

Date: Feb 2/2016

Time Allowed: 3:00hrs

Final Exam

Max. Mark 50%

General Instructions:

- Attempt All The Questions Carefully.
- Neatness Deserves Weightage.
- The Exam Is Closed Book.
- Assume Any Missing Data.

Name _____

ID.NO _____

6571

To be returned along with answer sheet

1. Discuss the following briefly:

- a. The basic objective of flood frequency analysis and its practical application. (3pts)
- b. The importance of flood routing and reservoir capacity determination. (3pts)

2. Explain the following terms

- a. Annual series and partial duration series (2pts)
- b. Recurrence interval (2pts)

3. Discuss the impact of sediment on the useful life of different hydraulic structures and its various methods which are adopted to control the deposition of sediment in reservoir. (3pts)

4. The record of annual peak discharge at a stream gauging station is as follows

year	1980	1981	1982	1983	1984	1985	1986	1987	1988
Q(m ³ /s)	311	720	430	330	248	125	405	400	233
year	1989	1990	1991	1992					
Q(m ³ /s)	178	248	321	186					

Determine

- a. The magnitude of a 10 year flood by using Log Pearson type III distribution methods. (5pts)
- b. The confidence limit of 95% by using Gumbel estimate for flood discharge of 10 years. (8pts)
- c. Reliability of the project by assuming its expected life is 25 years and designed for a flood magnitude of return period of 50 years. (2pts)

5. For a river reach $k=8h$ and $x=0.25$, route the following inflow hydrograph. (5pts)

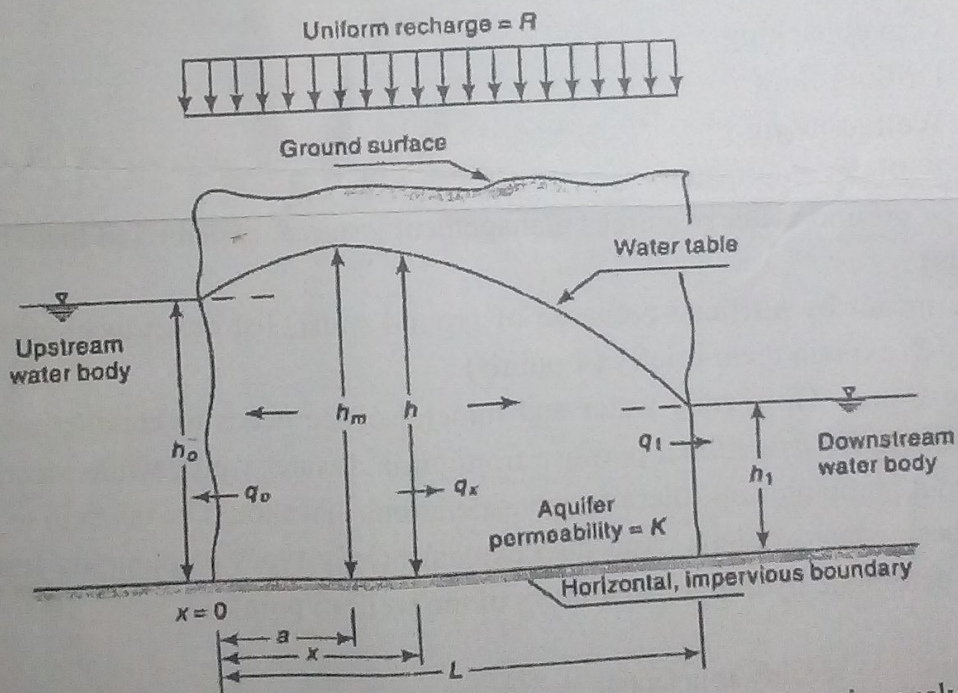
Time(h)	0	4	8	12	16	20	24	28
Inflow(m ³ /s)	8	16	30	30	25	20	15	10

The initial outflow discharge from the reach is 8m³/s.

Part Two (Work out)

1. An unconfined aquifer ($K = 5\text{m/day}$) situated on the top of a horizontal impervious layer connects two parallel water bodies X and Y which are 1200 m apart. The water surface elevations of X and Y measured above the horizontal impervious bed are 10.00 m and 8.00 m. If a uniform recharge at the rate of $0.002\text{m}^3/\text{day}/\text{m}^2$ of horizontal area occurs on the ground surface, estimate: (12 points)

- The water table profile.
- The location and elevation of the water table divide.
- The seepage discharges into the lakes.
- The recharge rate at which the water table divides coincides with the upstream edge of the aquifer.
- The total seepage flow per unit width of the aquifer at this recharge rate.



2. Artesian aquifer having $T = 600\text{m}^2/\text{day}$, $S = 0.0004$. The well is continuously pumped at the rate of $60\text{m}^3/\text{hr}$. assuming a well loss coefficient of $0.5\text{min}^2/\text{m}^5$, and index $n=2.3$, determine (5 points)

- specific capacity and
- The drawdown in the pumping well after 1 year of drought.

$R_w = 30\text{cm}$

ARBA MINCH UNIVERSITY
INSTITUTE OF TECHNOLOGY
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Final Examination

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 Examiner: Hydraulic str. team
 General instruction:

Section: G2HWRE:-A-H
 Date: June/2012
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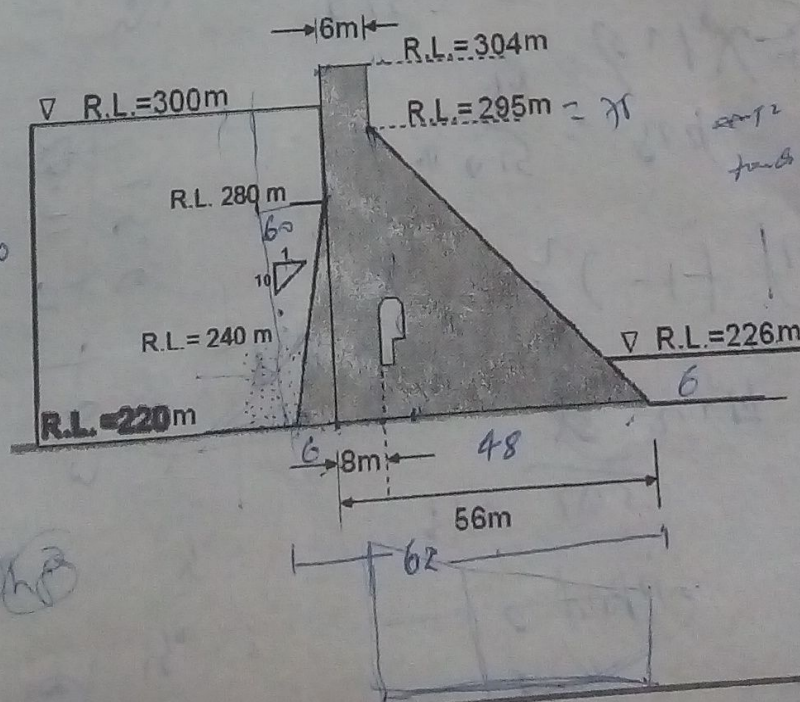
- ☞ Attempt all the questions carefully
- ☞ Neatness deserves weightage
- ☞ The exam is closed book
- ☞ Assume any missing data reasonably

1. A) Differentiate between concrete and embankment dam based on type, load transmission, topography and geology (site condition), construction material and mode of spillway construction. (5pts)

B) What is the difference between convention concrete and Roller compacted concrete, RCC? What are the different kinds of roller compacted concrete? What is the difference between these types? What are the conditions in which we choose one type from the other? (5pts)

2. The non-overflow gravity dam was supposed to construct at the coastal area of low lands. The dam site is subjected to high wind speed with the average wind velocity of 167.08Km/hr. and due to placement of the dam the impoundment will extended to 21Km. The site is supposed to be seismic free. Calculate; The stability against overturning and sliding. Use shear friction factor F_{SF} . (15pts) (only for G2HWRE-B,C,D,E,F)

Assume unit weight of concrete = 23.5 KN/m^3 , $C = 435 \text{ KN/m}^2$, $\phi_b = 32^\circ$, $\phi_s = 25^\circ$, $\gamma_{silt} = 19.3 \text{ KN/m}^3$ and allowable stress in concrete may be taken 2500 KN/m^2 .



ARBAMINCH UNIVERSITY

Department of Water Supply and Environmental Engineering

Final Examination

Subject: Water Supply & Treatment

Code: WSEE-3171

Section: G₃HWRE (C&D)

Max. Marks= 50%

Time allowed: 3 hours

*assume reasonably if any data is missed

PART-I

Answer the following questions correctly

1. What is the principal purpose of water treatment? (2 marks)
2. Define jar test (2 marks)
3. Show all the unit processes of a given water treatment plant with the help of sketch & define three of them (6 marks)
4. Mention three factors affecting the coagulation process (2 marks)
5. Define break down storage (2 marks)
6. How many types of distribution layouts are there? Discuss each of them with the help of sketch (6 marks)

PART II

Say True or False (1mark each)

7. It is possible to consider each individual substance of concern with the view to treatment.
8. Discharge & head loss are constant throughout a combined pipe flow system.
9. Taste & odor problems can be solved by applying aeration.

PART III

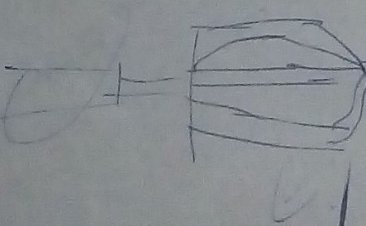
Choose the best answer (1 mark each)

10. Which state of chlorine is best for chlorination? ~~liquid~~
A. Liquid B. solid C. gas D. all
11. Which one of the following pipes has not capability of external load resistance?
A. Cast iron B. plastic pipe C. steel pipe D. ~~plastic pipe~~

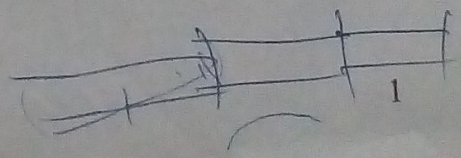
PART IV

Design properly

12. For the following pipe flow system, determine the reasonable flows in each pipe. (Neglect the minor losses & only two iterations for each loop are enough) (15 marks)



pipe	AB	BC	CD	DE	EF	FA	BE
Diameter (mm)	300	150	200	300	250	250	250
Length (m)	500	500	470	300	200	450	500
f	0.017	0.02	0.015	0.017	0.02	0.015	0.015



Arbaminch University
Institute Of Technology
Hydraulic And Water Resources Engineering Department

Course Name: Engineering Hydrology

Course Code: HENG-3143

Examiner: Assefa.G & Wondafrash A.

Section: G3HWRE-SEC A, B, C &D

Date: Feb 2/2016

Time Allowed: 3:00hrs

Final Exam

Max. Mark 50%

General Instructions:

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6571

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